



Anti-Sleep System

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DEDICATION

We dedicate this work to our family for their continuous support and encouragement, and to my professors who guided us throughout this journey

ACKNOWLEDGMENT

It is a profound honor to acknowledge the individuals who made this work possible. We deeply indebted to my supervisor, Adnan Al-Rabea'a, for his tireless support and expertise. His vision and technical direction helped navigate the complexities of this project, transforming challenges into milestones.

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DECLARATION

We hereby certify that this material, which we now submit for assessment on the programme of study leading to the award of Bachelor of Science in CS,SE,CIS is entirely our work, that we have exercised reasonable care to ensure that the work is original, and does not to the best of our knowledge breach any law of copyright, and has not been taken from the work of others save and to the extent that such work has been cited and acknowledged within the text of our work.

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ABSTRACT

Driver fatigue remains a critical global challenge, contributing to thousands of devastating traffic accidents annually.

Many drivers succumb to exhaustion without realizing the onset of "microsleep" until a catastrophic event occurs. To combat this issue, this project presents the design and development of an innovative, wearable "Anti-Drowsiness Eyewear" system. This intelligent device is engineered to detect driver fatigue in real-time, providing immediate sensory alerts and initiating a simulated emergency vehicle stop to prevent collisions.

The core architecture is built around the high-performance Arduino Uno microcontroller, integrating an infrared (IR) sensor, a high-frequency audible buzzer, and a precision vibration motor, all ergonomically mounted on a standard eyeglass frame.

The operational logic utilizes an IR sensor to continuously monitor eyelid movement through infrared light reflection.

By implementing a sophisticated two-second temporal threshold, the system effectively distinguishes between involuntary natural blinking and prolonged eye closure indicative of sleep.

Upon detecting fatigue, the microcontroller activates a dual-alert mechanism—both audible and tactile—to instantly rouse the driver. Simultaneously, the system triggers a relay module to simulate a controlled vehicle shutdown. Powered by a high-capacity 3.7V rechargeable lithium-ion battery, this device offers a lightweight, cost-effective, and highly reliable solution that fundamentally enhances road safety standards.

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LIST OF ABBREVIATIONS

ABBREVIATION	Definition of Abbreviation
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CHAPTER ONE

1 INTRODUCTION

1.1 PROJECT OVERVIEW

Drowsy driving stands as a primary contributor to catastrophic road accidents on a global scale. Scientific evidence suggests that even a brief interval of "microsleep" at high velocities can lead to a total loss of vehicle control, often resulting in fatal consequences. To mitigate this critical safety risk, the "Anti-Sleep Glasses" project introduces an innovative, wearable monitoring solution engineered for real-time fatigue, detection. The system is architected around the Arduino UNO microcontroller, serving as the central processing unit that interfaces with advanced infrared sensing technology to monitor eyelid patterns. Upon detecting prolonged eye closure—indicative of a transition into sleep—the device immediately activates a multi-layered alert mechanism consisting of a high-decibel buzzer and a precision vibration motor to rouse the driver. Furthermore, the system is integrated with an automated intervention circuit designed to simulate an emergency vehicle stop by deactivating the motor. This comprehensive approach not only alerts the operator but also provides a mechanical safety net to ensure the protection of both the driver and the public.

1.2 PROBLEM STATEMENT

Fatigue and drowsiness represent a significant threat to road safety, drastically impairing a driver's reaction time, situational awareness, and cognitive decision-making capabilities. Scientific research indicates that driving while sleep-deprived

can be as dangerous as driving under the influence of alcohol, as it often leads to "microsleep"—uncontrolled bursts of sleep that last only a few seconds but are sufficient to cause a total loss of vehicle control at high speeds. Despite the continuous advancements in automotive technology, a vast majority of vehicles currently in operation lack integrated, real-time drowsiness detection systems. Furthermore, the high-end safety solutions available in luxury vehicles are often prohibitively expensive and inaccessible to the general public. Consequently, there is an urgent and critical need for an affordable, non-intrusive, and highly reliable wearable system. Such a device must be capable of detecting the earliest physiological signs of exhaustion, providing immediate multi-sensory alerts, and initiating mechanical intervention to prevent catastrophic accidents and safeguard human lives.

1.3 PROPOSED SOLUTION

The proposed solution involves a wearable system equipped with an infrared (IR) obstacle avoidance sensor strategically mounted on an eyeglass frame to monitor the driver's eyelid activity. When the sensor detects prolonged closure, it triggers the Arduino uno to initiate a multi-layered response: an audible buzzer and a vibration motor for immediate arousal, and a 1-channel relay module to cut power to a dc gear motor (TT Motor). This combination of sensory feedback and mechanical shutdown provides a robust safety mechanism for long-distance driving

1.4 PROJECT OBJECTIVES

- **To design and implement an anti-sleep device using Arduino UNO.**
- **To detect eye closure accurately using an IR Obstacle Avoidance Sensor.**
- **To provide immediate auditory and tactile feedback (Buzzer and Vibrator) to the driver upon detecting drowsiness.**
- **To simulate vehicle intervention by stopping a DC motor using a Relay Module when the driver falls asleep.**

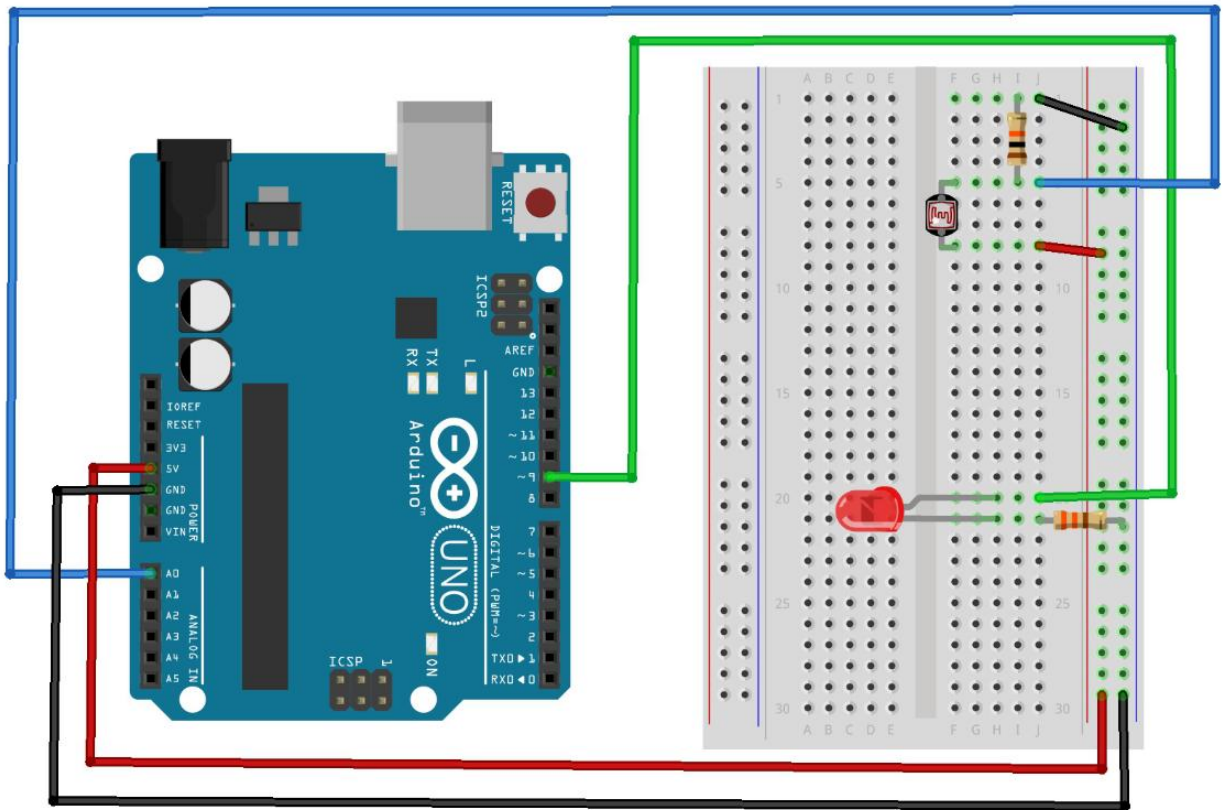
CHAPTER TWO

2 SYSTEM HARDWARE COMPONENTS

This chapter details the exact hardware components used in the project.

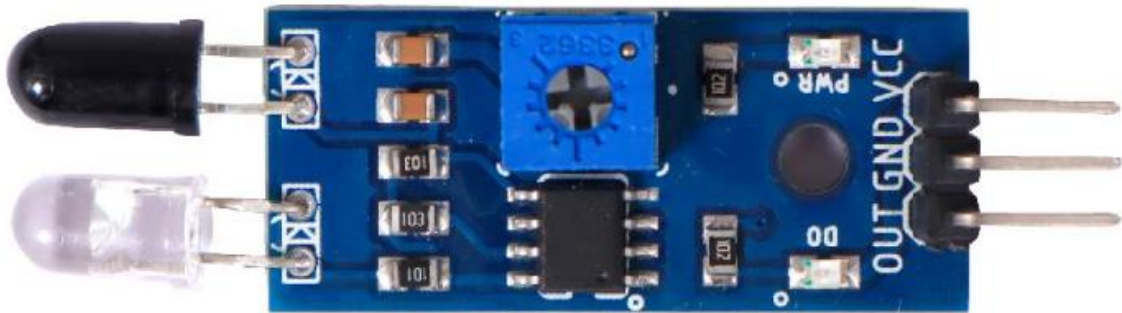
1.5 ARDUINO UNO AND BREADBOARD

The Arduino UNO serves as the main processing unit. A Mini Breadboard is used to connect the various modules and sensors securely using jumper cables.



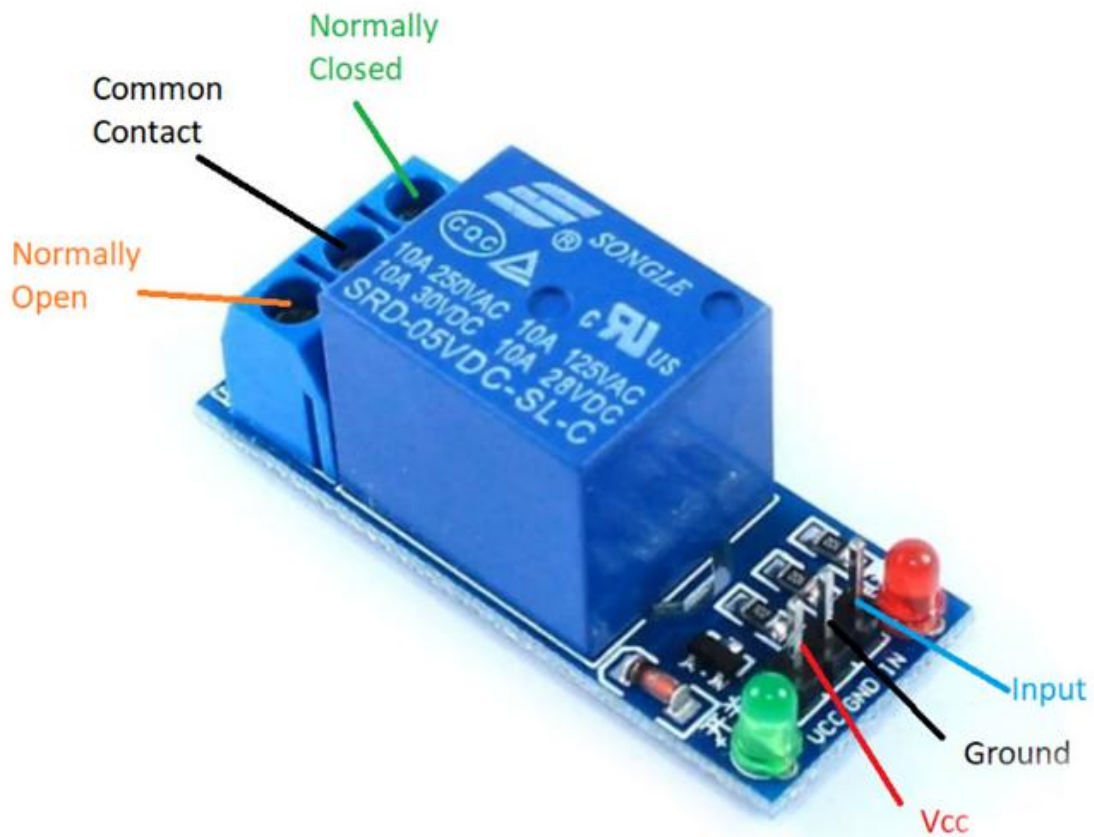
1.6 IR OBSTACLE AVOIDANCE SENSOR

The IR sensor continuously monitors the eye. It is connected to Digital Pin 2. When the eye is open, the sensor outputs a HIGH signal. When the eye closes, it outputs a LOW signal, triggering the alert system.



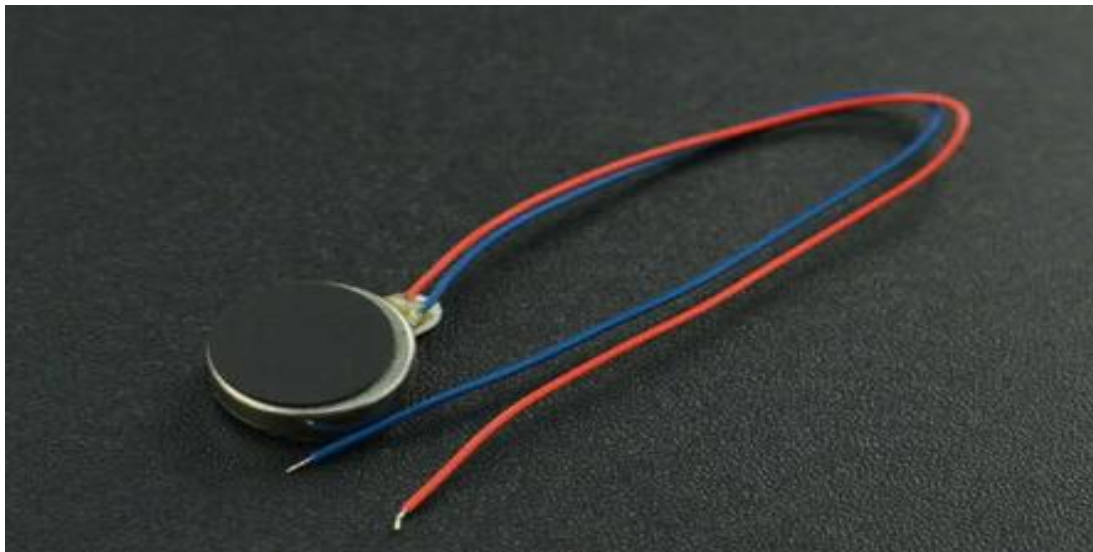
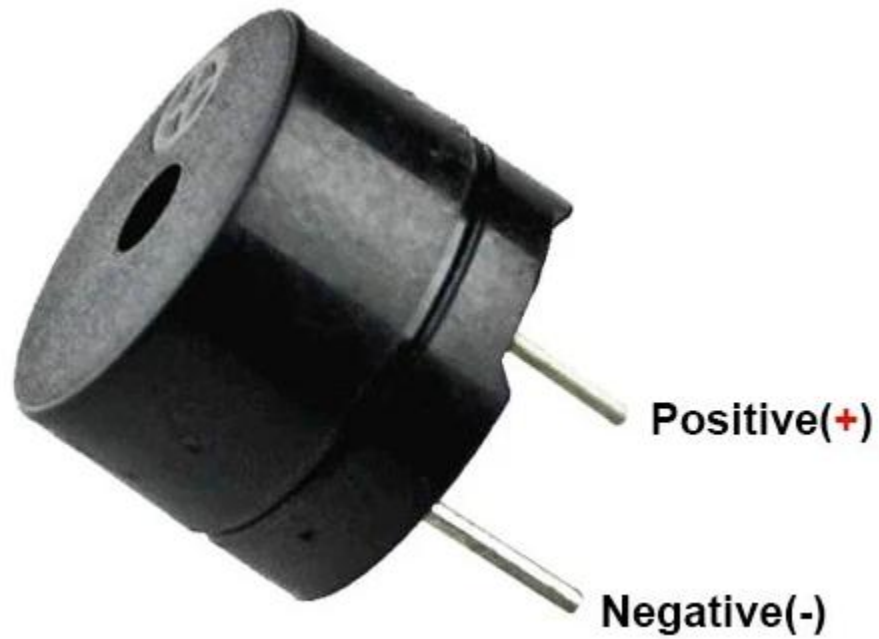
1.7 1-CHANNEL RELAY MODULE

The relay module acts as an electronically controlled switch for the vehicle's motor. It is controlled via Digital Pin 5. Under normal conditions, the relay allows power to flow to the motor. When danger is detected, it cuts the power.



1.8 ALERT MECHANISMS (BUZZER AND VIBRATOR)

- Active Buzzer: **Connected to Digital Pin 8, it provides a loud audible alert to wake the driver.**
- Vibrator: Provides tactile feedback to physically alert the driver.



1.9 CAR MOVEMENT SIMULATION COMPONENTS

To simulate a real vehicle, the following components are integrated:

- DC Gear Motor / TT Motor and Wheel: **Simulates the car's wheel**



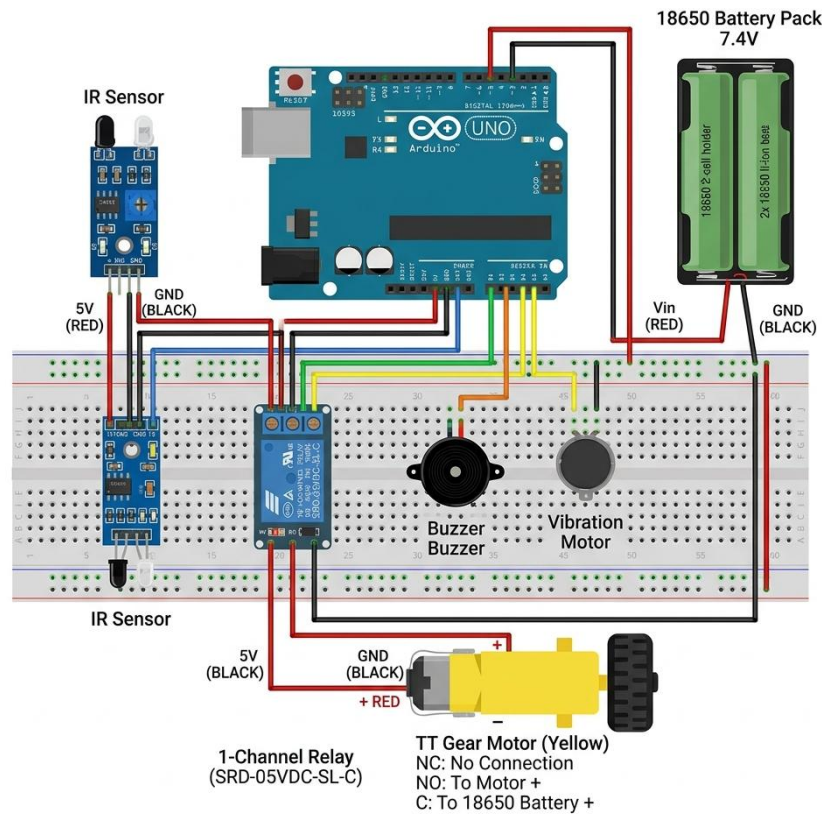
- Lithium-Ion Battery (18650) and Battery Holder: Provides a dedicated, high-capacity power source for the DC motor, separate from the Arduino logic power.



CHAPTER THREE

CIRCUIT DESIGN AND INTEGRATION

1.10 SYSTEM ARCHITECTURE IMAGE



1.11 PHYSICAL CONNECTIONS

The hardware integration follows a strict methodology to ensure safety and common grounding.

1. IR Obstacle Sensor (Input)

- VCC: Connects to the 5V rail on the breadboard.
- GND: Connects to the GND rail.
- OUT: Connects to Digital Pin 2 on the Arduino.

2. Relay Module (Output Control)

- VCC: Connects to the 5V rail.
- GND: Connects to the GND rail.
- IN: Connects to Digital Pin 5 on the Arduino.
- Load Side: The positive wire from the 18650 Battery goes to COM. The motor connects to the NC (Normally Closed) terminal.

3. Alarm System (Buzzer and Vibrator)

- Positive (+): Connects to Digital Pin 8 on the Arduino.
- Negative (-): Connects to the GND rail.

4. Power Supply (18650 Battery)

- Battery Positive (+): Connects to the Relay's COM terminal.
- **Battery Negative (-): Connects to the breadboard GND rail (Common Ground with Arduino).**

1.12 LOGICAL FLOW (METHODOLOGY)

- **Detection:** The IR sensor monitors the reflection from the eye.
- **Processing:** The Arduino reads the state of Pin 2. A LOW signal indicates the eye is closed (danger state).
- **Action:** The Arduino sets Pin 5 (Relay) to LOW to stop the motor, and Pin 8 (Buzzer/Vibrator) to HIGH to wake the driver.

CHAPTER FOUR

4 Software Implementation

1.13 4.1 ARDUINO SOURCE CODE

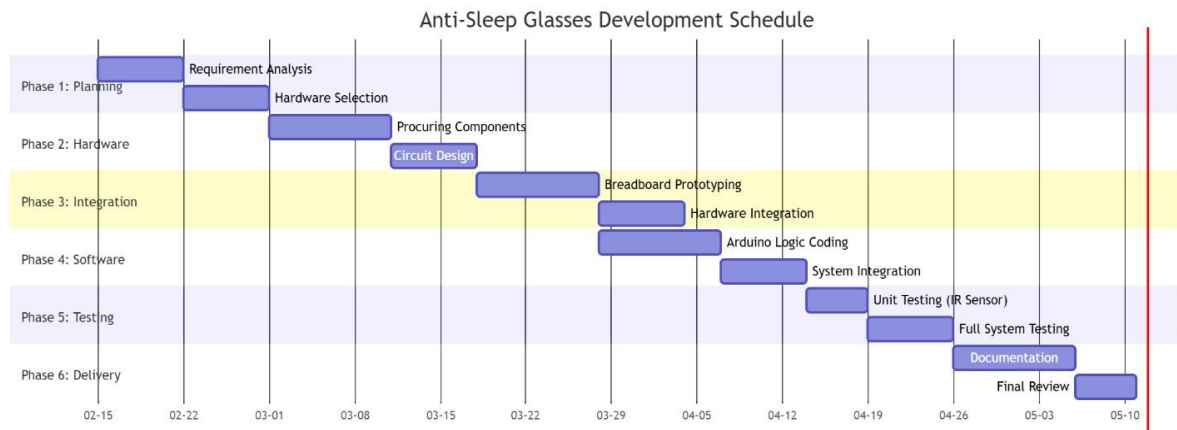
1.14 5.1 SYSTEM TESTING

The system was tested to ensure the IR Obstacle Avoidance Sensor correctly triggers upon eye closure. Upon detecting a closed eye, the relay module successfully cuts power to the 18650 Lithium-Ion battery, causing the TT motor to stop instantly while the active buzzer and vibrator emit a strong alarm.

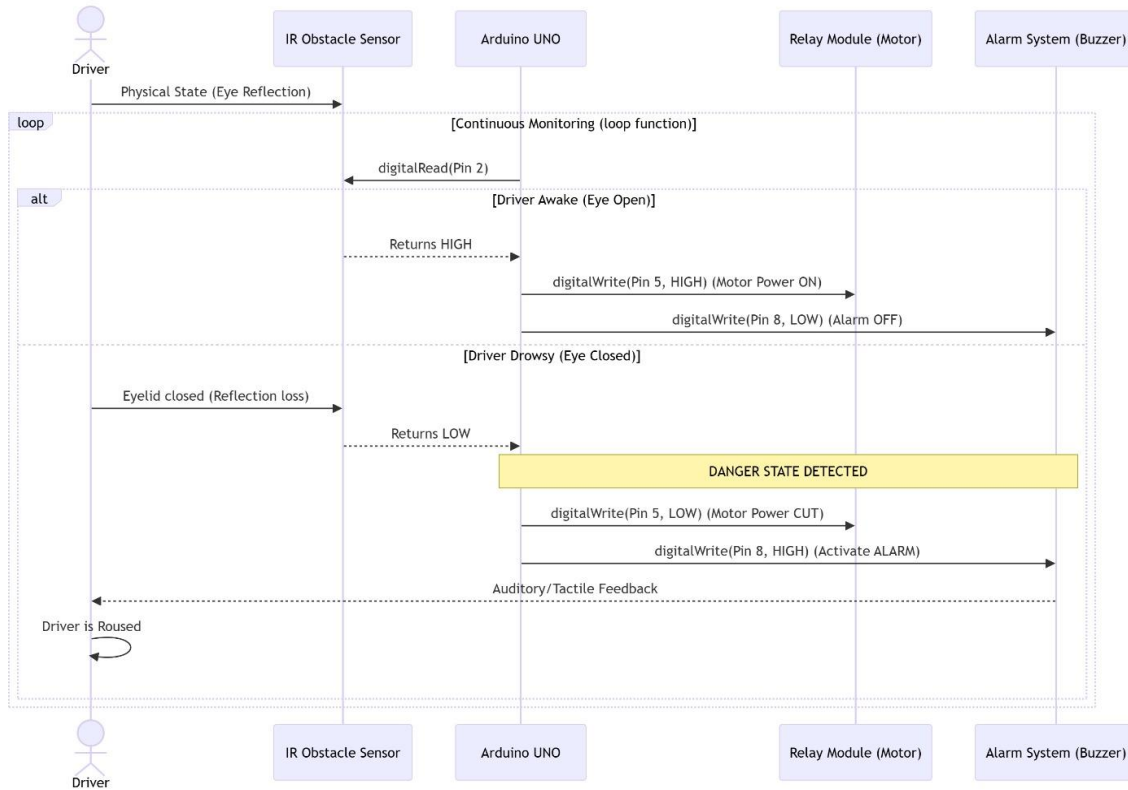
1.15 5.2 CONCLUSION

The integration of an Arduino UNO, an IR sensor, and a relay-controlled DC motor successfully demonstrates a functional prototype for drowsy driving prevention. The project simulates a real-world scenario where a vehicle can automatically halt if the driver is incapacitated.

Gant chart



UML sequence diagram



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